

A Mini Project Report on
AI Doctor with Vision and Voice

Submitted in partial fulfilment for the
degree of Bachelor of Technology in
Computer Engineering



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Subject – **Artificial Intelligence**

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2024-2025

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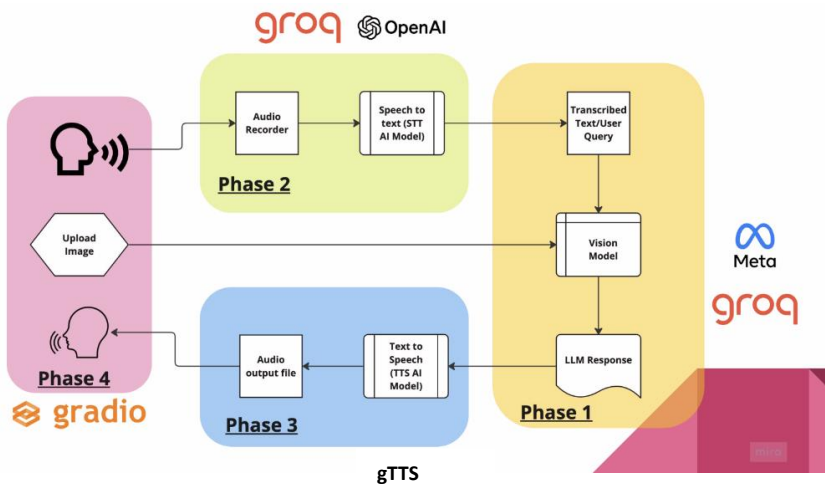
Objective

The AI Doctor with Vision and Voice - Medical Chatbot is a multimodal healthcare assistant that integrates voice, vision, and language understanding to simulate preliminary medical consultations. It combines speech recognition, computer vision, and large language models (LLMs) to process spoken symptoms and medical images and provide intelligent diagnostic feedback via text and voice.

Tools and Technologies Used

Component	Technology/Model	Purpose
Programming Language	Python	Core project logic
Speech-to-Text	OpenAI Whisper	Converts user voice to text
Large Language Model	Meta Llama-4-Scout-17B -16E-Instruct (via GROQ API)	Multimodal text and image understanding
Text-to-Speech	gTTS (Google Text-to-Speech)	Audio reply generation
Frontend Interface	Gradio	User-friendly interactive UI
Backend API	GROQ API	Model inference and conversational backend
Image Processing	OpenCV and Vision Model	Image analysis and interpretation

System Architecture



The system follows a **modular three-phase pipeline**:

- Speech and Vision Input Processing**
 - User uploads a **medical image** or records **voice input**.
 - Whisper converts voice into text transcriptions.
 - Images are processed through **Llama-4-Scout-17B-16E-Instruct**, capable of multimodal inference.
- Diagnosis Generation**
 - The Groq-powered Llama 4 Scout model processes both text and image data.
 - It outputs a **preliminary medical diagnosis or advice**.
- Output and Interaction**
 - The textual response is synthesized into audio using **gTTS**.
 - Gradio simultaneously displays text and plays the voice output, yielding a conversational experience.

Implementation Steps

1. Environment Setup

Python 3.10+, pipenv or venv for dependency management.

Libraries: gradio, openai-whisper, opencv-python, gtts, requests, and groq.

2. API Integration (GROQ)

Generated free GROQ API key for inference requests.

3. Voice and Vision Inputs

Implemented Whisper transcription pipeline.

Added image uploader linked to Llama4 model.

4. Text-to-Speech Integration

Used gTTS for free TTS generation (gTTS(text, lang='en').save('output.mp3')).

Integrated playback in Gradio app directly after inference.

5. Debugging and Runtime Fixes

Resolved Groq deprecation warning (adjusted endpoints).

Managed asynchronous response rendering in Gradio for smooth playback.

Model

llama-4-scout-17b-16e-instruct

The Llama-4-Scout-17B-16E-Instruct model is a state-of-the-art multimodal large language model from Meta, built on a Mixture-of-Experts (MoE) architecture with 17 billion active parameters and 16 expert networks.

Architecture

- **Base Transformer Core:** At its foundation is a standard transformer decoder architecture specialized for generating human-like text.
- **Mixture-of-Experts (MoE) Layers:** These layers contain 16 expert networks, but only a few are activated dynamically for each token. This selective activation enables the model to leverage a total 109 billion parameter pool efficiently.
- **Early Modality Fusion:** The model natively supports multimodal inputs, including text and images. These inputs are fused early in the pipeline before reaching transformer layers.
- **Token Routing:** A gating mechanism routes tokens to appropriate experts based on their context and content, balancing computational load.
- **Long Context Support:** The architecture supports extremely long contexts, handling up to 10 million tokens, which facilitates processing large documents or extended conversations.

Working

- The model takes in text and optionally image inputs, encoding images with a dedicated encoder.
- Inputs are fused in the early stages, producing combined multimodal token embeddings.
- Tokens are passed through the transformer decoder layers with mixture-of-expert routing, selectively engaging expert networks.
- The token outputs are autoregressively generated, allowing the model to perform text generation, question answering, summarization, code generation, and visual reasoning.
- The model is fine-tuned with instruction data for enhanced interaction, making it responsive to human commands and queries.
- It supports multilingual tasks and complex reasoning capabilities for versatile applications.

Use Case Highlights

- Multimodal assistant capable of handling chats involving images and text.
- Efficient large-scale document analysis and summarization.
- Visual question answering and image captioning.
- Tool use and coding assistance.

OpenAI Whisper

Type: Speech-to-text automatic speech recognition (ASR) model.

Architecture:

- Converts raw audio waveforms into log-mel spectrograms.
- Utilizes convolutional layers to extract acoustic features.
- Transformer-based decoder predicts transcribed text from audio features across multiple languages and accents.

Role: Converts patient's voice inputs into textual format for further processing by Llama 3 Vision.

Google Text-to-Speech (gTTS)

Type: Neural Text-to-Speech system.

Architecture:

- Text normalization and linguistic feature extraction.
- Acoustic model generates mel-spectrograms from normalized text.
- WaveNet vocoder synthesizes human-like audio from spectrograms.

Role: Converts textual diagnostic responses back to audio, enabling voice feedback to users.

Architecture

Fig 1 representing the multimodal AI workflow:

- Image encoder converts raw visuals into embeddings.
- Text encoder integrates linguistic context.
- Cross-attention fusion merges modalities.
- Transformer MoE layers generate reasoning-driven output.

This enables **visual question answering, captioning, and medical interpretation** in a single unified model.

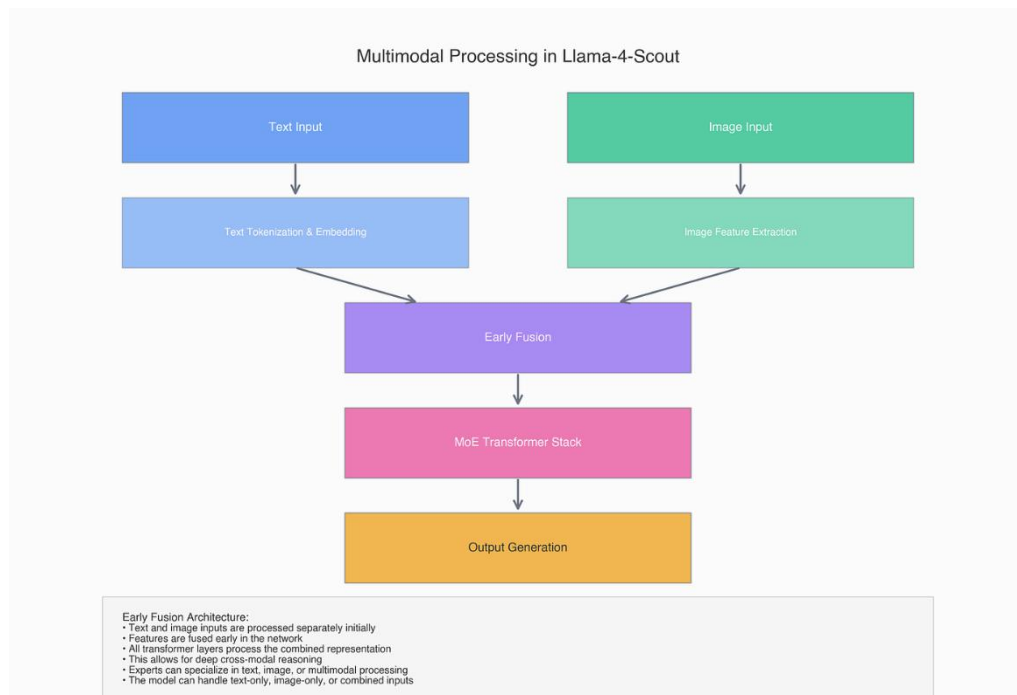


Figure 1 llama-4-scout-17b-16e-instruct model architecture

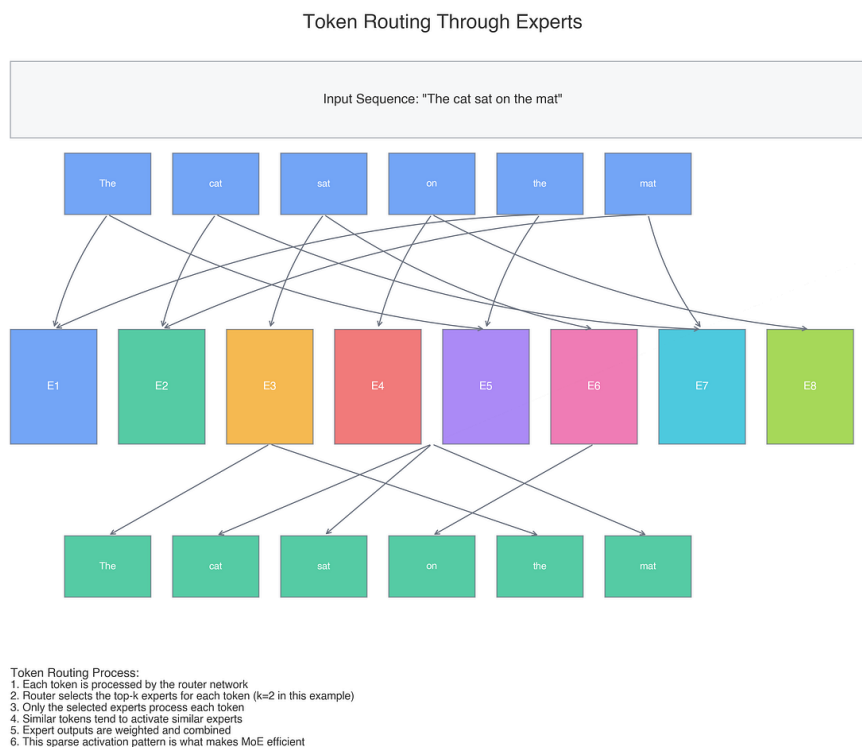


Figure 2 token routing in MoE

Fig 2 represents Mixture-of-Experts (MoE) tokenization and routing:

1. **Token Routing Concept** – Each token (text, image, etc.) is routed to a small subset of specialized *expert subnetworks* rather than processed by the entire model.
2. **Trainable Router (Gate)** – A learned gating network scores each token against all experts to decide which experts should process it.
3. **Top-k Expert Selection** – Only the top-k experts (with highest scores) are activated per token, enabling **conditional computation** and reducing computational cost.
4. **Weighted Expert Outputs** – The outputs from the selected experts are weighted (based on gating scores) and combined to update each token's representation.
5. **Efficiency and Scalability** – Since only part of the model is active per token, computation per token decreases while total model capacity scales up.
6. **Capacity Limits** – Each expert has a fixed capacity (number of tokens it can handle per batch) to avoid overload and maintain throughput.
7. **Load Balancing Losses** – Auxiliary losses encourage even token distribution among experts to prevent some experts from being overused.
8. **Routing Strategies** – Variants like *top-k routing*, *expert choice routing*, or *switch routing* determine how tokens are assigned to experts.
9. **Integrated Tokenization** – Tokenization and expert routing are intertwined, meaning token assignment directly influences how computation and specialization are distributed.
10. **Result** – The MoE mechanism achieves **efficient large-scale inference**, maintaining high capacity while keeping computational cost manageable.

Output

AI Doctor with Vision and Voice

Record your question

0:00

0:07

1x

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Upload Image



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Clear

Submit

Speech to Text

My hand is burning. Please tell me. I am getting a burning sensation in my hand. What to do?

Doctor's Response

With what I see, I think you have a skin lesion that appears to be a burn or possibly a skin reaction, and I would be concerned about the possibility of a second-degree burn or contact dermatitis given the redness and the shiny appearance of the skin. I would recommend applying a topical antibiotic ointment and a dressing to protect the area, and if the burning sensation persists or worsens, you should seek medical attention to rule out infection or other complications.

Doctor's Voice

0:35

0:35

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Flag

Use via API

Built with Gradio

Settings

32°C Sunny

Search

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ENG IN 🌐 📶 📶 📶 12:37 PM 17-10-2025

audio_filepath

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image_filepath



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Clear

Submit

Speech to Text

Doctor, what has happened to my foot? It is painful too much. Please let me know.

Doctor's Response

I'm truly sorry to hear that you're experiencing foot pain. With what I see, I think you have a possible case of ankle sprain or tendonitis. An ankle sprain occurs when the ligaments connecting the ankle joint become strained or torn, typically resulting from a sudden twisting motion. Tendonitis is inflammation of the tendons, which can be caused by overuse or repetitive strain on the Achilles tendon in the ankle. If you're experiencing pain, swelling, and limited mobility in your ankle joint, it's essential to consult a doctor or orthopedic specialist for a proper diagnosis and treatment plan. They may recommend rest, ice, compression, and elevation (RICE) to reduce pain and swelling, as well as physical therapy to improve mobility and strength. In some cases, surgery may be necessary to repair damaged ligaments or tendons.

output 2

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Flag

Use via API

Built with Gradio

Settings

AI Doctor with Vision and Voice

audio_filepath

0:00

0:07

1x

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image_filepath



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Clear

Submit

Speech to Text

Something on my screen. Please let me know what it is doctor. Please.

Doctor's Response

With what I see, I think you have a red rash or skin irritation on your arm. From a medical perspective, it could be due to various reasons such as an allergic reaction, infection, or skin condition like eczema or psoriasis. It's important to consult a doctor for a proper diagnosis and treatment.

output 2

🎵

Flag

Challenges Faced

- Component alignment in UI layout using Gradio.
- Managing asynchronous response rendering.
- Integration of audio playback post-inference.
- Fine-tuning API response timing for real-time interaction.

Errors and Scope of Improvement

Errors

- Lack of input validation in user interface.
- Limited test dataset for diagnosis scenarios.

Scope of Improvement

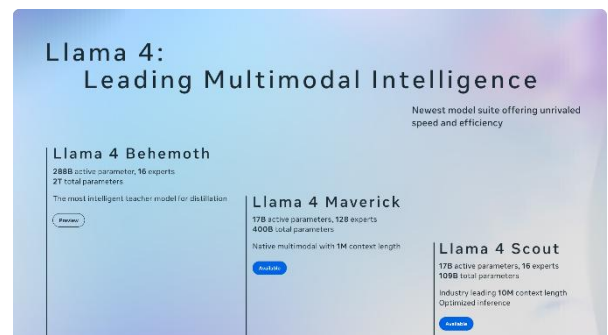
- Integration of dynamic medical datasets.
- Enhanced voice synthesis with neural TTS (e.g., ElevenLabs).
- Improved multilingual and noise-robust speech recognition.
- On-device deployment using quantized model variants.

Database and Knowledge Source

- The chatbot relies on **pretrained parameters** of **Llama-4-Scout-17B-16E-Instruct** as its implicit medical knowledge base.
- It does **not** use a traditional database; all reasoning is performed using embedded contextual knowledge.
- **Whisper** and **gTTS** use large audio corpora for transcription and synthesis, respectively.
- No patient data is stored; all processing is transient and secure.

Future Improvements

- Add retrieval-based knowledge augmentation for dynamic medical updates.
- Incorporate patient history management and secure data handling.
- Enhance voice naturalness with advanced TTS engines.
- Expand multilingual support and noise-robust speech recognition.
- Optimize for on-device inference for wider accessibility.



Conclusion

The AI Medical Chatbot 2.0 demonstrates a sophisticated integration of multimodal AI models to enable accessible, voice and vision-enabled healthcare interactions. Through efficient use of Groq hardware for Llama 3 Vision inference and open-source tools like Whisper and gTTS, the project achieves high diagnostic accuracy with real-time responses, suitable for educational, clinical assistive, and telehealth applications.